

SECTION III

The Neuroscience of Trauma and Dissociation

CHAPTER 10

The Neurobiology of PTSD

THE NORMAL FEAR RESPONSE

Fear is a normal response to alarming and out-of-the-ordinary situations. When we encounter something threatening, we respond on two different levels: (1) first unconsciously, through neurocircuits in the amygdala, which give rise to an automatic response, and then (2) consciously, through cognitive systems, such as the PFC and hippocampus, which give awareness to the emotion of fear (LeDoux, 2015). In other words, *we first respond unconsciously to a threat, and then we feel it consciously as fear.*

Fear is then *extinguished* when the ventromedial portion of the PFC regulates or calms down amygdala firing, and the hippocampus provides a context in which both learning and extinction can occur (LeDoux, 2015). For instance, your brain may tell your body: “That was scary, but we’ve experienced something like that before, we’re safe, and we can handle this.” We perceive a threat, we react, we’re afraid, we process the fear, and then we recover. This can happen occasionally or several times a day, depending on the level of stress we encounter. Remember one of the central tenets of IFS: *All protective parts are organized around fear.* Asking a client, for example, “What are you *afraid* would happen if you didn’t drink alcohol, get anxious, or withdraw?” is a way to get to the heart of this fear.

IMPLICIT AND EXPLICIT MEMORY

As we explore the neurobiology of PTSD and dissociation, it’s important to look at how memories, as well as fear, are encoded and processed. There are two common ways that memory can get encoded: via implicit memory or explicit memory. *Implicit memory* typically gets encoded within the first 18 months of life. However, we can also encode implicit memory whenever something traumatic happens to us throughout our life. Implicit memories are not language-based but, rather, are stored as perceptions, emotions, and bodily sensations. Implicit memory primes us for future action, is unconscious, and lacks awareness that it comes from the past (Anderson, 2013). I believe that most exiles and some extreme reactive parts are rooted in implicit memory.

Explicit memory, on the other hand, begins to develop at two years of age, typically when we begin talking. It requires focused attention; is linear, factual, and conscious; and involves a sense of time and narrative (Anderson, 2013). The hippocampus is necessary for explicit memory to develop. In IFS, we enable our clients to become aware of the unconscious exiled parts of themselves, housed in implicit memory, and guide them through a process of witnessing, unloading, and coping with their painful pasts. In other words, we help them convert implicit memory to explicit memory, which I believe is the overarching goal of trauma therapy in general.

POST-TRAUMATIC STRESS DISORDER

We'll now explore what happens in the brain and the body when the threat is more extreme or involves repeated violations of a verbal, physical, or sexual nature that result in the development of PTSD.

There are three main areas of the nervous system that, when affected by trauma, result in neural network dysregulation and symptom production: (1) neuroendocrine (i.e., hormonal), (2) neurochemical (i.e., neurotransmitters), (3) and neuroanatomical (i.e., brain structures). It's constructive to examine each system to better understand how PTSD develops, keeping in mind that protective parts and wounds likely live in the mind and utilize or access these trauma-based networks to express themselves.

HORMONAL FACTORS

When a stressful event occurs, our body receives this information and transmits it via nerve cells to the thalamus. When the information is deemed emotionally relevant, it heads toward the amygdala, which unconsciously processes this information, resulting in a variety of different reactions. One such response involves the activation of the hypothalamic-pituitary-adrenal (HPA) axis, which secretes a series of hormones. First, the hypothalamus secretes something called corticotropin-releasing hormone (CRH), which, in turn, stimulates the release of adrenocorticotrophic-releasing hormone (ACTH) from the pituitary gland, which ultimately stimulates the release of glucocorticoids—more commonly known as *cortisol* (the main stress hormone in the body, released from the adrenal glands located on the top of each kidney).

Hormones Released by the HPA Axis



Cortisol is what helps us mobilize and mount a response when something dangerous happens to us. It can be incredibly useful, even lifesaving under certain circumstances, but in excess it can be quite destructive. In moments of crisis, cortisol increases emotional stability and mobilizes glucose or energy for the fight-or-flight response. It's also a potent anti-inflammatory and anti-allergy agent. However, it can be quite harmful to the body when we are under stress for extended periods and a surplus builds up. It's known to be toxic to the hippocampus, it can cause protein breakdown and muscle wasting, and it's a potent immunosuppressant, which can increase the risk of infections, medical illnesses, and certain forms of cancer. Therefore, while the right amount of cortisol can be helpful, it can be harmful or neurotoxic in excess.

Under normal conditions, the release of cortisol activates the sympathetic nervous system and generates a negative feedback loop that quiets down both the hypothalamus and the anterior pituitary gland. However, it has been shown that this feedback mechanism is defective in certain individuals (due to increased negative feedback sensitivity of the HPA axis), which ultimately results in decreased levels of cortisol. Research also suggests that having low levels of cortisol at the time of a traumatic event can be a risk factor for the development of PTSD (Sherin & Nemeroff, 2011). Therefore, cortisol is now being studied as a potential treatment for acute trauma to help patients consolidate and process traumatic memories in the service of preventing the development of PTSD.

CHEMICAL FACTORS

Just as the amygdala stimulates the release of hormones as a result of a threat, it also releases several chemicals or neurotransmitters. We'll briefly review them now and learn how certain psychiatric medications that correct neurotransmitter imbalances can be helpful to treat the abnormalities created by trauma.

First, consider serotonin, or 5HT as it's commonly referred to. It's produced in a part of the brain called the dorsal and median raphe. Individuals with PTSD are known to have reduced serotonin, and these diminished levels are associated with hypervigilance, increased aggression, impulsivity, and enhanced formation of intrusive memories (Rege & Graham, 2017). Serotonin also regulates norepinephrine and CRH, both of which are excitatory and increase due to stress.

It's widely known that selective serotonin reuptake inhibitors (SSRIs) are the treatment of choice for PTSD. Restoring serotonin levels to normal amounts is shown to have an overall calming effect on the nervous system, as well as the body. SSRIs have also demonstrated an ability to restore PFC activity.

Next, we'll examine norepinephrine, one of the principal chemicals involved in the stress response. In the brain, norepinephrine primarily comes from an area called the locus coeruleus. When norepinephrine gets released, it becomes widely distributed to various parts of the brain involved in PTSD, including the PFC, amygdala, hippocampus, hypothalamus, periaqueductal gray (a structure affiliated with the freeze response), and thalamus. Norepinephrine increases fear, encodes emotional memory, and enhances arousal and vigilance (Sherin & Nemeroff, 2011). When the sympathetic nervous system is activated during stress, the adrenal glands release norepinephrine, in addition to cortisol, throughout the body (LeDoux, 2015), which contributes to symptoms such as increased heart rate, tingling sensations in the limbs, sweating, and increased breathing.

Medications that decrease norepinephrine levels in the body, such as propranolol (a beta blocker), or in the brain, like clonidine (an alpha-2 agonist), are extremely useful for decreasing excitability and treating acute trauma.

Dopamine is another excitatory neurotransmitter, like norepinephrine, which contributes to the development of anxiety, as well as fear conditioning. There's some evidence supporting a genetic component to the way dopamine is metabolized in the body that can determine whether or not someone will develop PTSD (Rege & Graham, 2017).

Atypical neuroleptics—such as quetiapine, risperidone, or aripiprazole, to name a few—are helpful in low doses for the treatment of dissociation. They partially block dopamine and the N-methyl D-aspartate (NMDA) receptor, which is connected to learning and memory.

Glutamate is the primary excitatory neurotransmitter in the brain, and stress directly triggers its release. Glutamate activates the NMDA receptor and is thought to contribute to the consolidation of traumatic memory and play a central role in the development of dissociation. It is toxic in excess and likely contributes to the loss of nerve cells in the hippocampus and PFC of PTSD patients (Sherin & Nemeroff, 2011).

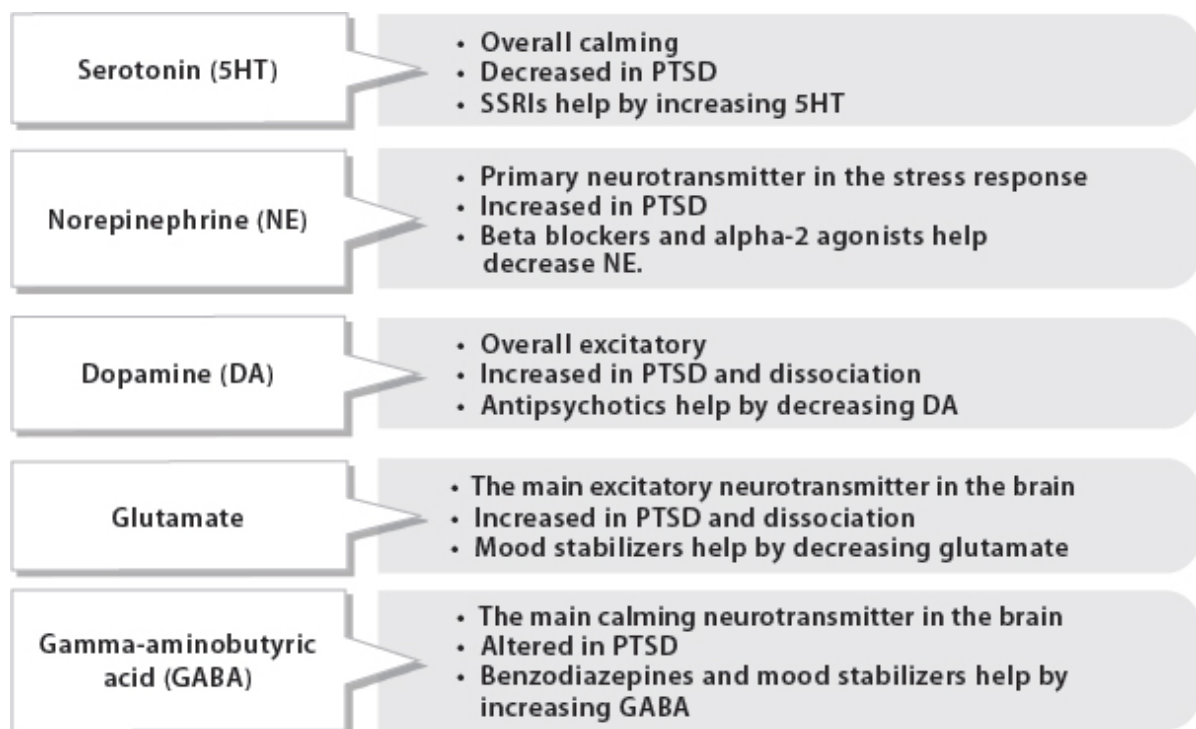
Mood stabilizers, which decrease glutamate, can be an excellent treatment for PTSD. However, many of them have numerous side effects, which make them difficult to tolerate.

GABA, on the other hand, is the primary inhibitory or calming neurotransmitter in the brain. It is closely connected to the benzodiazepine receptor, which drugs like lorazepam or clonazepam act upon. Stress is known to alter the GABA-benzodiazepine receptor complex (Sherin & Nemeroff,

2011), which is probably why so many people with PTSD seek out benzodiazepines to calm an activated brain.

Benzodiazepines are effective in treating the anxiety and activation associated with PTSD. Unfortunately, they can interfere with memory consolidation and are addictive, so they should be used sparingly or only on an as-needed basis. Certain mood stabilizers also increase GABA levels and are, therefore, calming to the nervous system.

The following chart summarizes the chemicals associated with the development of PTSD, along with medications that can help provide some relief.



ANATOMICAL FACTORS

Neuroimaging studies have confirmed some of the key structural changes in the brains of PTSD patients. Among these are the hippocampus, amygdala, and PFC. The PFC includes the anterior cingulate cortex, insula, and orbitofrontal cortex (Sherin & Nemeroff, 2011).

The *hippocampus*, which helps reduce amygdala firing, is an important structure connected to learning, memory, and the elimination of fear. Reduced hippocampal volume is known to be a hallmark feature of PTSD. Cortisol and glutamate, both of which are excitatory, have been shown to impair new nerve growth and damage nerve cells in the hippocampus (Rege & Graham, 2017). Some researchers have speculated whether smaller hippocampal volume is a predisposing factor (or precursor) of PTSD development or a result of the toxic effects of the trauma itself (Sherin & Nemeroff, 2011).

The *amygdala* is crucial in helping us process emotions and extinguish fear. With increased stress, the amygdala releases glutamate, which encodes traumatic memories via the NMDA receptor, as previously mentioned. Studies have shown amygdala hyperresponsiveness and increased reactivity in clients who suffer from PTSD (Sherin & Nemeroff, 2011).

The *ventromedial portion of the PFC* exhibits inhibitory control over the stress response and emotional reactivity by suppressing amygdala firing (Sherin & Nemeroff, 2011). Individuals with PTSD have smaller PFC and anterior cingulate cortex volumes as a result of the noxious effects of trauma, and this loss in volume is thought to be a result of trauma, not a potential preexisting condition, as some have postulated is the case with the hippocampus (Rege & Graham, 2017).

Hippocampus	Amygdala	Prefrontal Cortex
• Calms amygdala firing • Reduced volume in PTSD • Preexisting condition or a result of toxic stress?	• Increases glutamate and consolidates trauma memory • Increased excitability and reactivity as a result of PTSD	• Reduces fear by inhibiting the amygdala • Reduced volume of ventromedial prefrontal cortex and anterior cingulate cortex in PTSD

It's not surprising that stress can negatively affect parts of the brain that are essential for recovering from threat, fear, and violation. And when that recovery isn't possible, symptoms consistent with PTSD arise. Luckily, the medications discussed earlier are known to be helpful in reversing some of these changes.

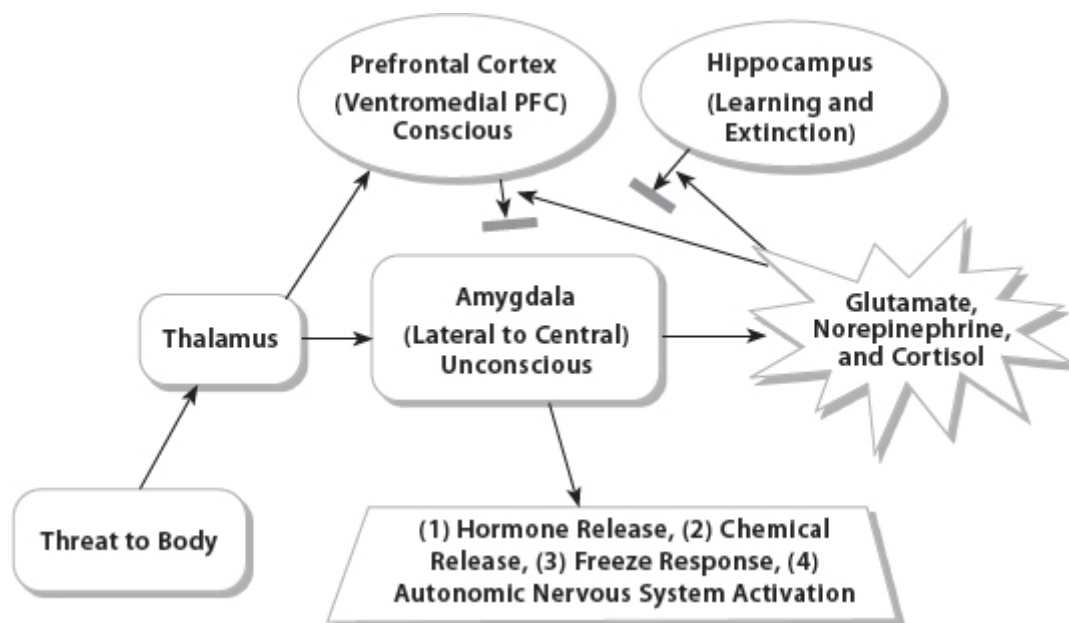
Even more remarkable, however, is IFS's ability to help clients access trauma-affected parts that are in some way associated with the hormonal, chemical, and structural changes in the brain and body. When we help our clients access Self-energy, they can unburden their wounds; release the thoughts, feelings, and sensations affiliated with their relational violation; and seemingly untangle and rewire their brains back to their original, non-traumatized states. This connection between IFS, trauma, and the brain is continually fascinating to me—a connection I hope we will continue to better understand and clarify through future research.

PUTTING IT ALL TOGETHER

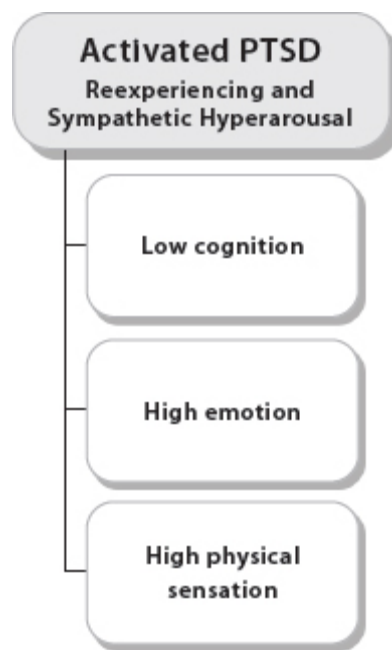
In summary, when we experience an intrusion, we try our best to counteract the assault and recover from the threat. Sensory input from the body enters the thalamus, where it then moves up to the PFC and amygdala. Remember: The amygdala is quick and unconscious, and the PFC is slower and

conscious in its processing of the information (van der Kolk, 2014). The amygdala responds by stimulating the release of hormones (CRH, ACTH, and cortisol) and chemicals (serotonin, acetylcholine, norepinephrine, and dopamine), activating the freeze response and regulating the sympathetic and parasympathetic nervous systems in an attempt to recover. The PFC and hippocampus then process the information and attempt to downregulate or calm down the activation in the amygdala (i.e., recovery or elimination).

When conditions are repeated, extreme, and void of love and connection, the amygdala releases glutamate (excitatory), which stimulates the release of large amounts of norepinephrine in the brain and body. Cortisol also continues to escalate until it eventually becomes depleted. These excitatory chemicals and hormones become toxic to the PFC and hippocampus, rendering them ineffectual in quieting the overwhelm, giving rise to the development of PTSD.

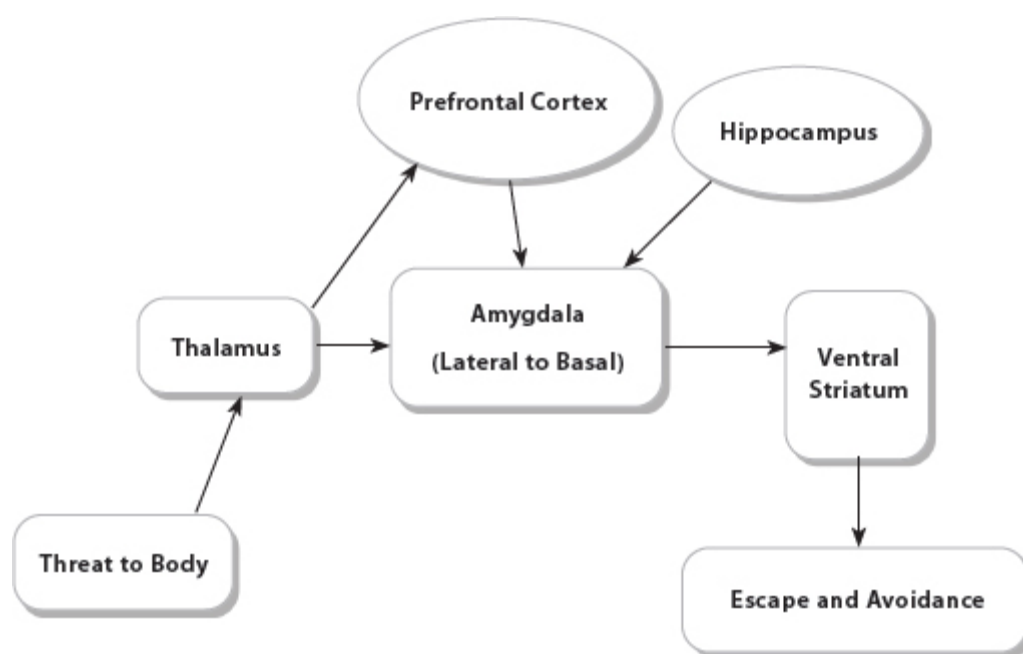


This excitatory response, shown in the diagram above, represents an “activated subtype” of PTSD described by Lanius and colleagues. Individuals with the activated subtype account for 70 percent of PTSD clients, and they exhibit reexperiencing and physiologic hyperarousal (Frewen & Lanius, 2015; Lanius et al., 2010). Neuroimaging studies have confirmed that these individuals exhibit low activation of the PFC and anterior cingulate cortex, which Lanius has called “failed inhibition” or the inability of the PFC to calm down or inhibit the amygdala from firing (Lanius et al., 2010). Here, emotions are high, the body is activated, and cognitions are low.



Now that we have discussed activated PTSD, take a moment to consider the other side of the story: when PTSD clients present with symptoms or parts that are blunted, numb, withdrawn, or dissociative. This type of blunted or dissociated PTSD accounts for 30 percent of clients (Frewen & Lanius, 2015), but it has been less thoroughly researched and is less understood compared to activated PTSD.

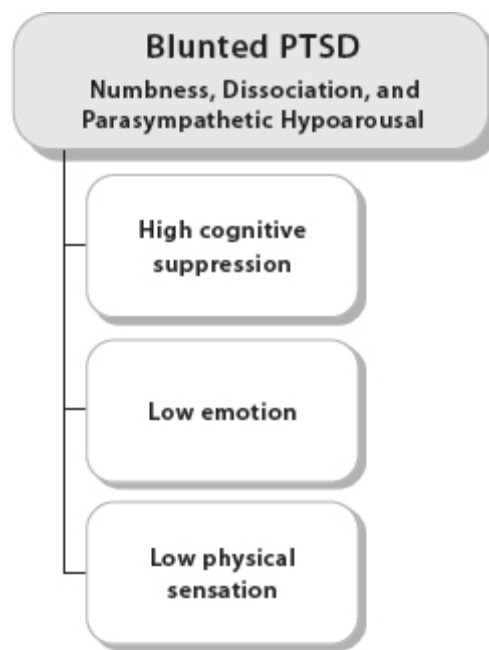
However, LeDoux and colleagues have been able to identify a shift within the amygdala that occurs with the avoidance response. Unlike the fear conditioning response, in which information travels from the lateral amygdala to the central amygdala, during the avoidance response, information goes from the lateral amygdala to the basal amygdala. *This redirected information inhibits the freeze response and activates a withdrawal reaction.* The basal amygdala connects with a structure in the brain called the ventral striatum (specifically the nucleus accumbens), which results in motor actions consistent with escape and avoidance (LeDoux, 2015). Here is a diagram showing the avoidance response.



Stephen Porges, a scientist at the Kinsey Institute and professor at the University of North Carolina, also helps us understand the avoidance, dissociation, numbing, and shutdown responses through his polyvagal theory. His theory focuses on the autonomic nervous system and the different ways we react and respond to safety, stress, and life threat.

Porges describes how our body constantly and unconsciously assesses our surrounding environment to determine whether or not we're safe through a process he calls *neuroception* (Porges, 2017). When we experience safety in connection, the polyvagal theory maintains that the *ventral vagal branch of the parasympathetic nervous system* is active. However, when we experience danger, our *sympathetic nervous system* activates, causing us to respond in the familiar fight-or-flight manner. If we perceive that this danger is a life threat from which there is no escape, the *dorsal vagal branch of the parasympathetic nervous system* then activates, and we consequently disconnect, shut down, and dissociate as a survival mechanism (Porges, 2017).

Lanius has also described what happens in the brain for chronic PTSD survivors who exhibit avoidance, numbing, and dissociative symptoms. Neuroimaging studies show that individuals with this type of blunted PTSD exhibit the exact opposite response of those with hyperarousal, who show decreased PFC activity. In contrast, those with blunted PTSD exhibit an *increased* reaction in the PFC and anterior cingulate cortex. Lanius calls this state “overmodulation” or over-inhibition of emotional responsiveness (Lanius et al., 2010). With blunted PTSD, cognitive suppression is high while emotionality and physical sensations are low.



In summary, Lanius and her group have described two different subtypes of PTSD: a reexperiencing and hyperaroused type (activated PTSD) and a dissociative and hypoaroused type (blunted PTSD; Frewen & Lanius, 2015). In IFS, we think of these subtypes as different extreme protective parts that can express themselves in an activated or hyperaroused state, while other parts can protect in a blunted, withdrawn, or dissociative manner.

CHAPTER 11

Dealing with Extreme Parts: How Neuroscience Informs Therapeutic Decisions

It's helpful to further investigate these two different subtypes of PTSD and extreme parts—hyperaroused and hypoaroused—that attempt to protect traumatic wounds.

The intense reactions from parts commonly associated with relational trauma are typically what makes working with these clients so challenging. The severity and zealous nature of these first responder-type parts can challenge the best and most skilled therapist. It's important to remember, especially when the reactivity and intensity are unwavering, *that a protective part might need to unload or unburden itself before it's willing to permit access to the wound*. Extreme parts often hold so much anguish and self-loathing that they may opt to first release their pain before healing the wound.

When we're in the presence of extremely protective parts, we often don't have the luxury of being able to stop, to take a break, and to identify what's getting activated within ourselves before responding to them. As therapists, we need to do our best in the moment not to overreact—to hold the space for our client's activated parts and to assist them in feeling heard, seen, and validated for the ways they are trying to help. These parts often sabotage treatment, block progress, and detract from accessing internal pain and vulnerability. Neuroscience knowledge can aid therapists in making strategic therapeutic decisions that overcome these common roadblocks and help extreme parts more effectively unblend and give access.

When I think about extreme parts, I first like to focus on the *energy* they possess and present in the treatment room. For example, an alcohol-drinking part may say: "I'm so done with him. Give me the damn vodka now. I'm so pissed off that I could just kill him." The energy here is reactive and sympathetically hyperaroused. Compare that to another alcohol-drinking part presenting with energy in the resigned realm, which may say: "I'm done. What's the point? I give up. Give me the vodka—I'm out of here." Here the energy is more withdrawn, hopeless, and parasympathetically hypoaroused. The point is that parts can present with high or low energy, and it's important to differentiate between the two because each requires a different neuroscience-informed IFS response from the therapist.

Once you've determined the energy of your client's part, it's necessary to notice your own parts that may be present. Ask yourself: "How do I respond internally to the energy of my client's part?" Perhaps you find yourself thinking something like: "I'm sick of her coming in week after week complaining about that person and resorting to drinking as her only coping mechanism." This is an angry and activated response from one of your parts. Or you might be thinking: "My mind keeps wandering. This session feels like it's going on forever. Will this client ever get better?" This is a bored, distracted, and defeatist response from one of your parts.

It's important to identify our activated parts and ask them to give us space so we can be fully present for our clients' parts. The therapist's Self would respond with compassion, openness, and curiosity.

ACTIVATED AND HYPERAROUSSED PARTS

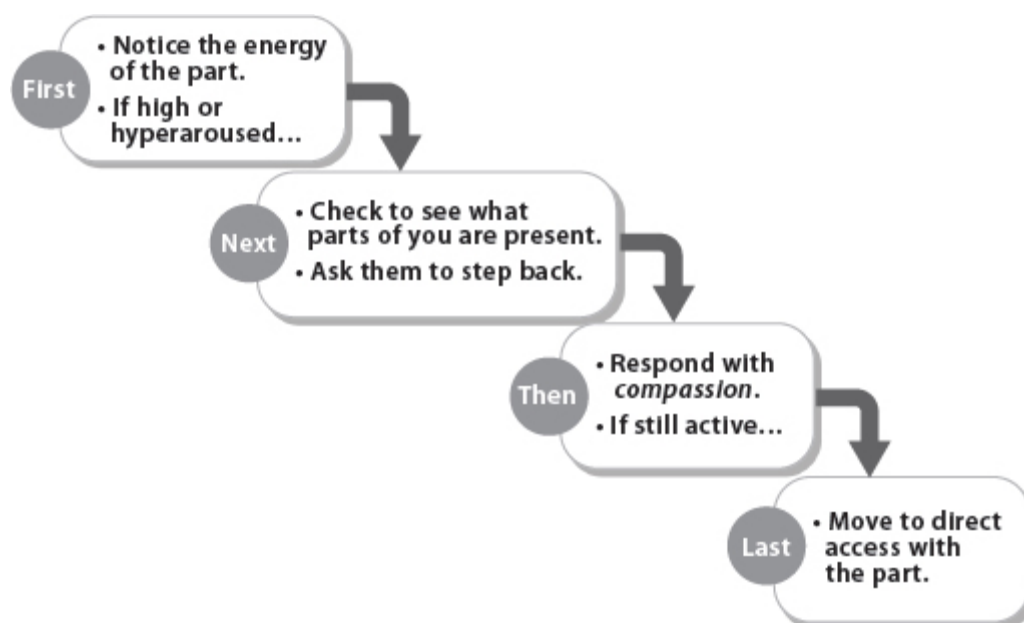
When clients are blended with hyperaroused or high-energy parts, remember that neuroscience tells us they have little or no access to their PFC in these moments. This results in high emotion and high physical sensation, with little or no capacity for objectivity and reasoning. *Here, it's best to respond to them with compassion, not empathy.* Remember that compassion is caring about the other person's suffering with the desire to help. It has distance and perspective, is "unblended" in IFS terms, and is consistent with Self-energy. When your client's brain is temporarily unavailable to them, you can offer the separation, care, and perspective they need and are unable to provide for themselves.

You bring compassion to the part when you say things like: "You seem angry right now. But I know you'll be able to get through this. I'm here and available to help you in any way I can." This type of response is more likely to help the part relax and unblend.

If, on the other hand, you are more empathically resonate (i.e., feeling distress similar to your client), you run the risk of fueling the client's activation. This can happen when you offer a remark such as: "You look anxious right now. I know what that feels like. I feel a bit anxious right now too." Don't forget that emotions beget emotions.

When you're with a client with activated parts, try your best to maintain a caring, compassionate, and unblended stance. If this doesn't help them separate and their parts remain blended, I recommend moving to *direct access*. Lend them your PFC; in other words, have your Self engage in a dialogue with their aroused part, provided you have access to yours at that moment. Your Self can talk directly to their part, which is especially helpful when your client's brain is temporarily taken over and offline.

In summary, with activated, extreme parts, take the following steps:



NUMB AND DISSOCIATED PARTS

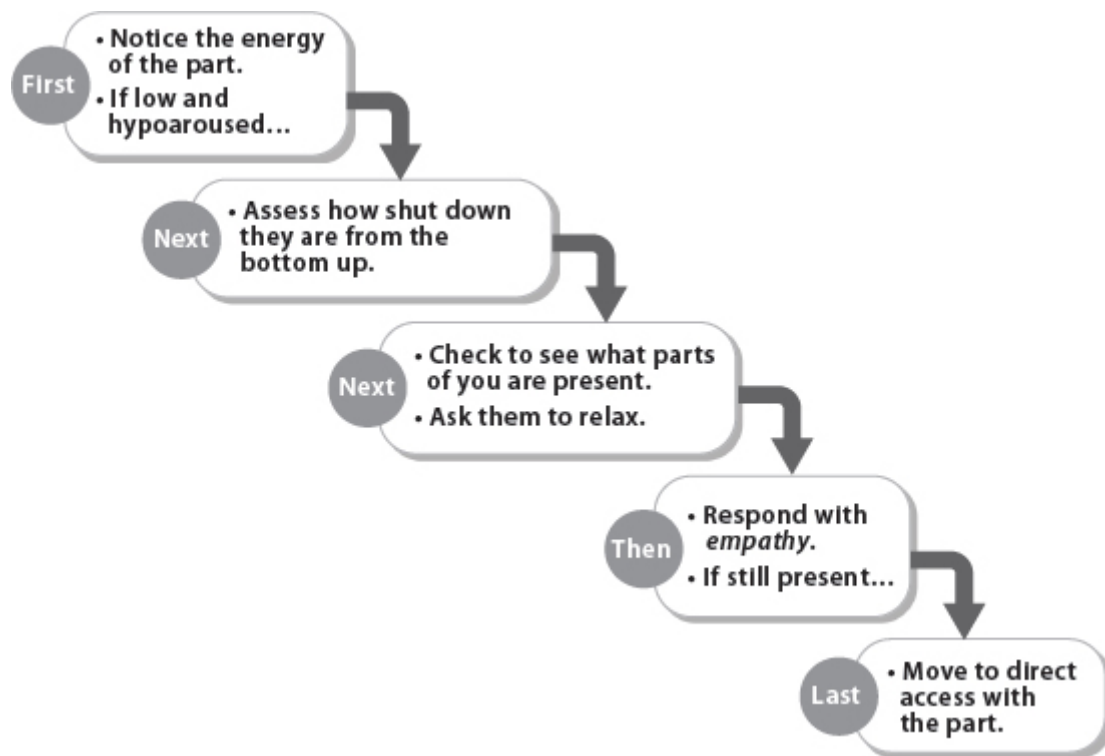
When the energy of the part is blunted, shamed, or withdrawn, clients can often internally experience this energy as being more problematic than hyperaroused energy. Porges reminds us that when we perceive a life threat, the dorsal branch of the vagus nerve activates. It's also well-known that it takes longer to recover from a state of shutdown than it does to recover from the fight-or-flight response. Additionally, we know that when we're withdrawn, the PFC is overactive (versus being underactive during arousal) and shuts everything down below it from the top down. This results in little or no access to thoughts, feelings, or physical sensations.

When you're with a client's part that presents with slow, low, or disconnected energy, I suggest that you first evaluate the degree to which their system is shut down at that moment. I might say something to the client like: "Can you wiggle your fingers right now? Are you feeling anything? What are you thinking about right now?" I assess how disconnected the client is from the bottom up—meaning body first, then feelings, followed by thoughts. Neuroscience teaches us that over-inhibition is a top-down process, so evaluating from the bottom up makes the most sense.

Once I've determined how disconnected the client is, I'll check in with myself to see if any of my parts are present. If not, I'll try to help the part unblend, this time with empathy instead of compassion. *Empathic resonance* is more likely to help the client bring their shutdown system back online. I would say: "I'm with you right now. I sense how upsetting that was for you. I know what it's like to be shamed. It really feels horrible. I can relate."

Again, if my empathic connection doesn't help the client's part unblend, I move into *direct access* using my Self-energy to connect directly with the client's shutdown part.

Try these steps:



Below is a summary of neuroscience-informed interventions with extreme parts. It's important to note that with more severe trauma, such as DID, the parts that are present can rapidly alternate. It can be challenging to determine which part is present in a particular moment and what energy it's carrying. Having a clear game plan when encountering extreme traumatized parts is beneficial for therapists and clients alike. It enables clients' parts to relax and to unblend more quickly and minimizes the likelihood that therapists' parts will get activated.

